MIL-L-55574A(EL)
25 November 1975
SUPERSEDING
MIL-L-55574(EL)
15 February 1968

MILITARY SPECIFICATION

LAMP, XENON, SHORT-ARC, 10/15/20 KW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of Short-Arc, Xenon Lamp for use in the 30 inch general purpose searchlight, AN/TVS-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5855

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MILITARY

MIL-P-116	Preservation-Packaging Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-M-13231	Marking of Electronic Items
MIL-C-46842	Cushioning Material, Unicellular Polyethylene Foam (for Packaging Purposes)
MIL-S-55572	Searchlight, Infrared, AN/TVS-3
MIL-I-55573	Ignitor, Xenon Lamp for Searchlight, Infrared, AN/TVS-3

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

DRAWINGS

ELECTRONICS COMMAND

SC-D-599425	Arc Lamp, 20 KW, Xenon
SC-D-599426	Lamp and Container Assembly
SC-C-599872	Assembly, Container and Coolant

(Copies of specification, standards, and drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

Standard Welding Symbols AWS.A2 0-58

(Application for copies should be addressed to American Welding Society, 435 East 47th Street, New York, N. Y. 10016)

NATIONAL BUREAU OF STANDARDS

Handbook H-28

(Application for copies should be addressed to National Bureau of Standards, Springfield, Virginia 22151)

3. REQUIREMENTS

3.1 Description. The lamp shall be a short-arc lamp utilizing xenon gas in a quartz envelope and shall have liquid cooled electrodes. The lamp shall operate in any attitude. The lamp shall be fitted with locating and supporting rings and coolant line connecting fittings which also serve as power connections.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article lamps in accordance with 4.3.

3.3 Construction. The lamp shall be constructed in accordance with SC-D-599425 and as specified herein.

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provision of this specification and shall conform to MIL-P-11268.

3.5 Major components. The following components shall be constructed in accordance with applicable drawings, specifications and as specified herein.

3.5.1 Quartz bulb. The quartz bulb shall be constructed in accordance with drawing SC-D-599425 and as specified herein.

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3.5.2 Xenon gas. The lamps shall be filled with xenon gas of not less than 99.8 percent purity to a pressure of 2.9 to 3.3 atmospheres at plus 24°C, plus or minus 2°C.

3.6 Operational and environmental.

3.6.1 Optical quality. Optical quality of the quartz bulb shall be such that the apparent shift of the cathode observed in any direction shall lie within a circle of 1.5 millimeter radius. The bulb shall be free from imperfections such as bubbles, scratches, and visible distortions in the entire area between the support rings.

3.6.2 Lamp ignition. The lamp shall produce a visible arc within 1 second and shall achieve full sustained arc within 2 seconds after application of ignition impact power from an igniter conforming to MIL-I-55573.

3.6.3 Operating power level. The lamp, after ignition, shall operate continuously at each of the following power levels:

a. 10 kW at 34.8 volts dc, plus or minus 1.5 volts dc.

b. 15 kW at 39.3 volts dc, plus or minus 1.7 volts dc.

c. 20 kW at 43 volts dc, plus or minus 1.8 volts dc.

3.6.4 Starting current. The lamp shall withstand, without damage, a starting current of 825 amperes, plus or minus 25 amperes for 1 second, plus or minus 100 milliseconds duration in each 5 minutes of operation.

3.6.5 Mechanical shock. The lamp shall not be damaged (see 6.3) by mechanical shocks of 15 g's (see 6.3) with each shock having a duration of 6 milliseconds, plus or minus 0.5 millisecond.

3.6.6 Vibration. The lamp shall not be damaged by vibration throughout a frequency range of 15 to 200 cycles per second at a level of 2.5 g's, plus or minus 0.5 g.

3.6.7 Low temperature operation and storage. The lamp shall perform as specified herein at an ambient temperature of minus 45°C and shall not be damaged by storage in its shipping container at a temperature of minus 56°C for extended periods with humidity uncontrolled.

3.6.8 High temperature operation and storage. The lamp shall perform as specified herein at an ambient temperature of plus 177°C and shall not be damaged by storage in its shipping container at a temperature of plus 74°C for extended periods with humidity uncontrolled.

3.6.9 Altitude. The lamp shall perform as specified herein at any altitude from sea level to 8,000 feet above sea level. The lamp, in its shipping container shall not be damaged by transit in military aircraft, (unpressurized) at altitudes to 40,000 feet above sea level.

3.6.10 Lamp burn-in. The lamp shall perform as specified herein for a minimum of 12 hours continuous operation at the 20 kW power level and repeated intermittent operations.

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